

Numerical Project

Triangular Profile Fin

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I. Problem Statement & Geometry

The purpose of this project is to analyze a triangular profile fin that is being subjected to convection from the top and bottom surfaces. The goal is to determine the temperature profile along the fin using the finite difference numerical method and to evaluate the heat flow rate at the base and the heat flow rate lost by convection from the fin's surface. The results of this project will verify the energy balance and provide more information about the thermal performance of the fin.

Problem geometry

Since the problem has a triangular profile, the equations that are going to be used are:

$$y_u(x) = 0.2(L - x)$$

$$y_l(x) = -0.2(L - x)$$

The fin is constricted into 9 equally spaced nodes using the step size of 0.01 m, resulting in 8 control volumes.

Dimensions			
Parameters	Symbol	Value	Units
Base Temperature	T_0	115	°C
Surrounding Temperature	T_∞	15	°C
Heat transfer coefficient	h_c	100	$W/m^2 \cdot K$
Thermal conductivity	k	15	$W/m \cdot K$
Step size	Δx	0.01	m

Fin Length	L	0.08	m
Base thickness	t	0.032	m
Number of nodes		9	

Table 1

Assumptions

- Steady-state heat conduction.
- One-dimensional temperature variation along the fin length.
- Constant thermal conductivity k .
- Negligible heat loss from the side surfaces (wide fin).
- Uniform convection coefficient over the entire surface.
- Negligible radiation effects.

II. Finite - Difference Equations

Given a triangular fin that has a variable cross-sectional area and convection at its surface. The energy balance at the node using the finite volume technique will be shown below.

Energy balance

$$\frac{kA_{j+1/2}(T_{j+1}-T_j)}{\Delta x} - \frac{kA_{j-1/2}(T_{j-1}-T_j)}{\Delta x} - h_c P_j(T_j - T_\infty)\Delta x = 0$$

Where the cross-sectional area is: $A(x) = 0.4(L - x)$

And the perimeter for convection is: $P(x) = 2\sqrt{1 + 0.2^2}(L - x)$

Conduction from the left face: $\frac{kA_{j-1/2}(T_{j-1}-T_j)}{\Delta x}$

Conduction out to the right face: $\frac{kA_{j+1/2}(T_{j+1}-T_j)}{\Delta x}$

Convection loss from top and bottom surfaces: $h_c P_j (T_j - T_\infty) \Delta x$

Convection coefficient: $= hP(x) \Delta x$

Tip convection factor: $hP(\frac{\Delta x}{2})$

Number of nodes: $N = \frac{L}{\Delta x} + 1 = 9$

Node Equations

To start the iterations and set up the node equations, the following control volume equation was used:

$$\frac{kA_{j-1/2}(T_{j-1} - T_j)}{\Delta x} + \frac{kA_{j+1/2}(T_{j+1} - T_j)}{\Delta x} + hA_{conv,j}(T_\infty - T_j) = 0$$

This equation applies for each interior node, with the exception of the tip having a different equation

$$\frac{kA_{j-1/2}(T_{j-1} - T_j)}{\Delta x} + hA_{conv,j}(T_\infty - T_j) = 0$$

Node equations used in the code:

Node 0:

$$T_0 = 115$$

Node 1:

$$45(T_1 - T_0) + 39(T_1 - T_2) - 0.14277255(T_1 - 15) = 0$$

Node 2:

$$39(T_2 - T_1) + 33(T_2 - T_3) - 0.12237647(T_2 - 15) = 0$$

Node 3:

$$33(T_3 - T_2) + 27(T_3 - T_4) - 0.10198039(T_3 - 15) = 0$$

Node 4:

$$27(T_4 - T_3) + 21(T_4 - T_5) - 0.08158431(T_4 - 15) = 0$$

Node 5:

$$21(T_5 - T_4) + 15(T_5 - T_6) - 0.06118823(T_5 - 15) = 0$$

Node 6:

$$15(T_6 - T_5) + 9(T_6 - T_7) - 0.04079216(T_6 - 15) = 0$$

Node 7:

$$9(T7 - T6) + 3(T7 - T8) - 0.02039608(T7 - 15) = 0$$

Node 8(tip):

$$3(T8 - T7) - 0.000000000(T8 - 15) = 0$$

III. Numerical Implementation

Method Selection & Justification

Using the Gauss-Seidel iterative method to solve the system of equations makes it possible to update each node using the most recent values of the temperatures. It also starts from an initial guess and iteratively refines the solution. This method is simple and efficient. The Gauss-Seidel method is a good option for diagonally dominant systems because of the need to add the conduction and convection terms, like the triangular fin, avoiding the use of matrices. It allows us to control the convergence tolerance and observe how the iteration behaves. It is easy to apply and extend to more complicated boundary conditions.

Algorithm outline

Steps:

- I. Input all given data and set the problem
 - A. Constants in Table 1
 - B. Find number of nodes and positions
- II. Compute geometry quantities
 - A. Find the cross-sectional height, slant factor ($\sqrt{1 + 0.2^2}$)
 - B. Find the perimeter and convective areas (face and tip)
- III. Start Temperature
 - A. Set temperature array: $T_0 = 115^\circ C$

- B. Set initial guess for other nodes: $T_{\infty} = 15$
- IV. Loop with Gauss-Seidel (iterative solving)
 - A. For each node:
 - 1. Find conduction at both sides
 - 2. Calculate the convection
 - 3. Update T
 - B. For the tip node:
 - 1. Use left conduction and half-face convection
 - C. Remember to update temperature, track convergence, and set counters
 - D. Check the maximum change in T and set it to stop when it's below the tolerance
- V. Create a temperature table and plot
- VI. Verify energy balance and convergence
 - A. Find heat flow rate at the base
 - B. Find heat lost by convection
 - C. Compare to check energy balance

IV. Results

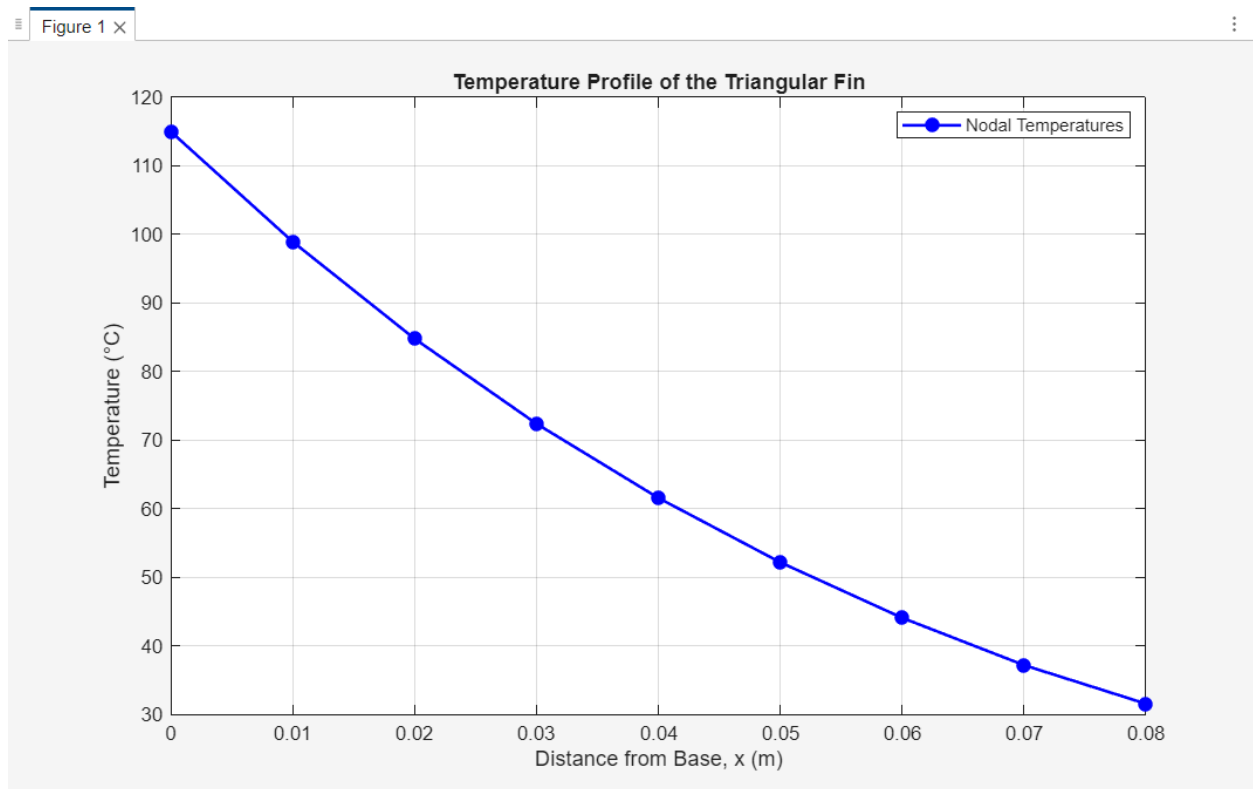
Temperature table

Temperatures	
Node	Temperature (°C)
T(0)	115.0000
T(1)	98.9372
T(2)	84.7928
T(3)	72.3904
T(4)	61.5673

Temperatures	
T(5)	52.1746
T(6)	44.0796
T(7)	37.1780
T(8)	31.5516

Table 3

Plot



Convergence Report

The final solution converged in 76 iterations. Meanwhile, the final maximum change was approximately 9.89×10^{-7} .

V. Calculations & Verification

Constants:

$$T_0 = 115 \text{ }^{\circ}\text{C}$$

$$T_{\infty} = 15 \text{ }^{\circ}\text{C}$$

$$h = 100 \text{ W/m}^2 \cdot \text{K}$$

$$k = 15 \text{ W/m} \cdot \text{K}$$

$$L = 0.08 \text{ m}$$

$$t = 0.032 \text{ m}$$

Conduction

Method 1:

Where the cross-sectional area is:

$$A(x) = 0.4(L - x)$$

$$A(0) = 0.4(0.08)$$

$$A(0) = 0.032 \text{ m}^2$$

And the perimeter for convection is:

$$P(x) = 2\sqrt{1 + 0.2^2}(L - x)$$

$$P(0) = 2\sqrt{1 + 0.2^2}(0.08)$$

$$P(0) = 0.1631686244 \text{ m}$$

$$m = \sqrt{((h)P(0))/(A(0)k)}$$

$$m = 5.830391361 \text{ m}^{-1}$$

$$\Theta(x) = T(x) - T_{\infty}$$

$$\Theta(0) = T(0) - T_{\infty}$$

$$\Theta(0) = 115 - 15$$

$$\Theta(0) = 100 \text{ }^{\circ}\text{C}$$

$$\Theta(0) = \Theta_b$$

$$Q = kA(x)m\Theta_b((\sinh(mL) + (h/mk)(\cosh(mL)))/(\cosh(mL) + (h/mk)(\sinh(mL))))$$

Method 2:

Geometry Functions:

$$A(x) = 0.4(L - x) = 0.032 \text{ m}^2$$

$$P(x) = 2\sqrt{1 + 0.2^2} (L - x)$$

Node temperatures:

Table 3

Heat flow rate at the base:

Face location for base: $x = dx/2 = 0.005\text{m}$

$$A(x=0.005) = 0.030\text{m}^2$$

$$Q_{\text{base_conduction}} = -kA_{\text{face}} \frac{T_1 - T_0}{\Delta x}$$

Heat leaving the base into the fin

Base convection

$$P(0) = 2s(L-0)$$

$$A_{\text{base_conv}} = P(0) * (dx/2)$$

$$Q_{\text{base_conv}} = hA_{\text{base_conv}} * (T_0 - T_{\text{inf}})$$

$$Q_{\text{total}} = Q_{\text{base_cond}} + Q_{\text{base_conv}}$$

Total convection loss

Heat lost by convection:

Interior node:

$$A_{\text{conv}} = 2*s*dx$$

$$Q_{\text{conv_total}} = q_{\text{conv_total}} + h_c * A_{\text{interior}} * (T(j) - T_{\text{inf}});$$

Tip node:

$$A_{\text{tip}} = 2*s*(dx/2.0)$$

$$Q_{\text{conv_total}} = q_{\text{conv_total}} + h_c * A_{\text{tip}} * (T(N) - T_{\text{inf}});$$

The analytical expression for fin heat transfer was used to compare against the numerical base heat flow. The percentage difference was small, confirming correct implementation. Minor deviations are attributed to discretization and modeling assumptions.

The equations used for conduction and convection are the same for the assumptions made in the project, but several factors may introduce uncertainty in the final heat transfer results. The triangular fin has a continuously changing cross-sectional area, and it is divided into 8 control volumes. This may create small numerical deviations from the analytical results. In addition to this, convection at the tip is treated as a half-surface area, meaning that any mismatch in how the tip convection is modeled vs the analytical assumptions may lead to an error in total heat loss.

VI. Analysis & Conclusion

Numerical choices

The Gauss-Siedel method was selected due to our familiarity with it from the Modeling Methods class. This method is characterized by its fast convergence and simplicity. A small convergence tolerance 10^{-6} ensured accuracy along with 76 iterations and 9 nodes.

Physical interpretation

The temperature decreased nonlinearly along the fin from 115 °C to 31.5 °C, base to tip. This follows the expected behavior of a triangular fin. This is because the cross-sectional area decreases, which reduces the fin's ability to conduct heat. The convective surface area remains high even as the cross-sectional area decreases.

Given:

$$T_0 = 115\text{ °C}$$

$$T_{\infty} = 15 \text{ }^{\circ}\text{C}$$

$$h = 100 \text{ W/m}^2 \cdot \text{K}$$

$$L = 0.08 \text{ m}$$

$$A(0) = 0.032 \text{ m}^2$$

$$P(0) = 0.1632$$

$$A_s = P(0) \cdot L = 0.01306$$

$$Q = 824.81$$

Fin Effectiveness:

$$\varepsilon = Q / (h(A(0))(T_0 - T_{\infty}))$$

$$\varepsilon = 2.58$$

Fin Efficiency

$$\eta = Q / (hA_s(T_0 - T_{\infty}))$$

$$\eta = 0.63$$

The fin effectiveness $\varepsilon = 2.58$, $2.58 > 2$. This indicates that the fin has an enhanced efficiency when compared to a plane with a consistent cross-sectional base area throughout. The efficiency of 63% shows a loss due to the temperature gradient along its length.

Conclusion

The approach we used correctly modeled the heat conduction and convection for the triangular fin with a varying cross-section. The results indicate that the fin gives sufficient thermal performance with good efficiency and effectiveness. This project shows the reliability of the Gauss method for analyzing complex geometries where analytical solutions are difficult to calculate.

VII. Appendix

Code

```
clear;
clc;
close all;
T0 = 115.0;      % Base temperature (deg C)
T_inf = 15.0;    % Surrounding fluid temperature (deg C)
hc = 100.0;      % Heat transfer coefficient (W/m^2*K)
k = 15.0;        % Thermal conductivity (W/m*K)
L = 0.08;        % Fin length (m)
dx = 0.01;       % Step size (m)
% --- 2. Discretization and Geometry ---
N = (L / dx) + 1; % Total number of nodes
x = linspace(0, L, N); % Node positions
% The slope is derived from y_u(x) = 0.2(L-x).
s_factor = sqrt(1 + 0.2^2); % Surface length factor
% Function for cross-sectional area (per unit width W=1)
Ac = @(x_val) 0.4 * (L - x_val);
% --- 3. Initialize Temperature Array ---
T = ones(1, N) * T_inf; % Initial guess: All nodes at T_inf
T(1) = T0;              % Set boundary condition at the base
% --- 4. Gauss-Seidel Iterative Solution ---
% This loop fulfills the iterative method requirement
tolerance = 1e-6;
max_iterations = 10000;
converged = false;
for iteration = 1:max_iterations
    T_old = T; % Store values from the previous iteration
    for j = 2:N-1
        % Calculate geometry at control volume faces
        Ac_west = Ac(x(j) - dx / 2.0);
        Ac_east = Ac(x(j) + dx / 2.0);
        As_conv = 2.0 * s_factor * dx; % Convection surface area
        % Calculate coefficients for the FDE
        C_west = k * Ac_west / dx;
        C_east = k * Ac_east / dx;
        C_conv = hc * As_conv;

        % In Gauss-Seidel
        T(j) = (C_west * T(j-1) + C_east * T_old(j+1) + C_conv *
T_inf) / ...
            (C_west + C_east + C_conv);
    end
    % --- Special case: Tip Node (j = N) ---
    j = N;
```

```

    % Geometry for tip node
    Ac_west = Ac(x(j) - dx / 2.0);
    As_conv = 2.0 * s_factor * (dx / 2.0);

    % Calculate coefficients
    C_west = k * Ac_west / dx;
    C_conv = hc * As_conv;
    % FDE for tip node: T(j) uses the newly calculated T(j-1)
    T(j) = (C_west * T(j-1) + C_conv * T_inf) / (C_west + C_conv);
    % --- Check for Convergence ---
    error = max(abs(T - T_old));
    if error < tolerance
        converged = true;
        break;
    end
end
% --- 5. Results (Part 2: Temperature Profile) ---
fprintf('--- Temperature Profile ---\n');
if converged
    fprintf('Solution converged in %d iterations.\n', iteration);
else
    fprintf('Solution did NOT converge after %d iterations.\n',
max_iterations);
end
fprintf('\nTemperature Profile:\n');
fprintf('Node | x (m) | Temperature (C)\n');
fprintf('-----|-----|-----\n');
for j = 1:N
    % T(j) is the temp, x(j) is position.
    fprintf(' %-3d | %6.3f | %15.4f\n', j-1, x(j), T(j));
end
% Plot the temperature profile
figure;
plot(x, T, 'bo-', 'LineWidth', 1.5, 'MarkerFaceColor', 'b');
title('Temperature Profile of the Triangular Fin');
xlabel('Distance from Base, x (m)');
ylabel('Temperature (°C)');
grid on;
legend('Nodal Temperatures', 'Location', 'northeast');
% --- 6. Results (Part 3: Heat Flow Rate) ---
fprintf('\n--- Heat Flow Rate ---\n');
% Method 1: Heat flow at the base
% q_base = q_conduction_out + q_convection_out
Ac_half = Ac(x(1) + dx / 2.0); % Area at x=dx/2
As_0 = 2.0 * s_factor * (dx / 2.0); % Surface area for node 1
q_cond_out = k * Ac_half * (T(1) - T(2)) / dx;
q_conv_0 = hc * As_0 * (T(1) - T_inf);
q_base = q_cond_out + q_conv_0;

```

```

fprintf('Heat flow rate at base: %.4f W/m\n', q_base);
% Method 2: Heat lost by convection from the total surface
q_conv_total = 0.0;
% Convection from node 1
q_conv_total = q_conv_total + hc * As_0 * (T(1) - T_inf);
% Convection from interior nodes
As_interior = 2.0 * s_factor * dx;
for j = 2:N-1
    q_conv_total = q_conv_total + hc * As_interior * (T(j) -
T_inf);
end
% Convection from tip node
As_tip = 2.0 * s_factor * (dx / 2.0);
q_conv_total = q_conv_total + hc * As_tip * (T(N) - T_inf);
fprintf('Heat lost by convection: %.4f W/m\n', q_conv_total);
% Check for equivalence
fprintf('Equivalence Check (Difference): %.4f W/m\n',
100*(abs(q_base - q_conv_total))/q_base);

```

Screenshot

```

--- Temperature Profile ---
Solution converged in 76 iterations.

```

Temperature Profile:

Node	x (m)	Temperature (C)
0	0.000	115.0000
1	0.010	98.9372
2	0.020	84.7928
3	0.030	72.3904
4	0.040	61.5673
5	0.050	52.1746
6	0.060	44.0796
7	0.070	37.1780
8	0.080	31.5516

```

--- Heat Flow Rate ---
Heat flow rate at base: 824.8085 W/m
Heat lost by convection: 824.8084 W/m
Equivalence Check (Difference): 0.0000 W/m

```